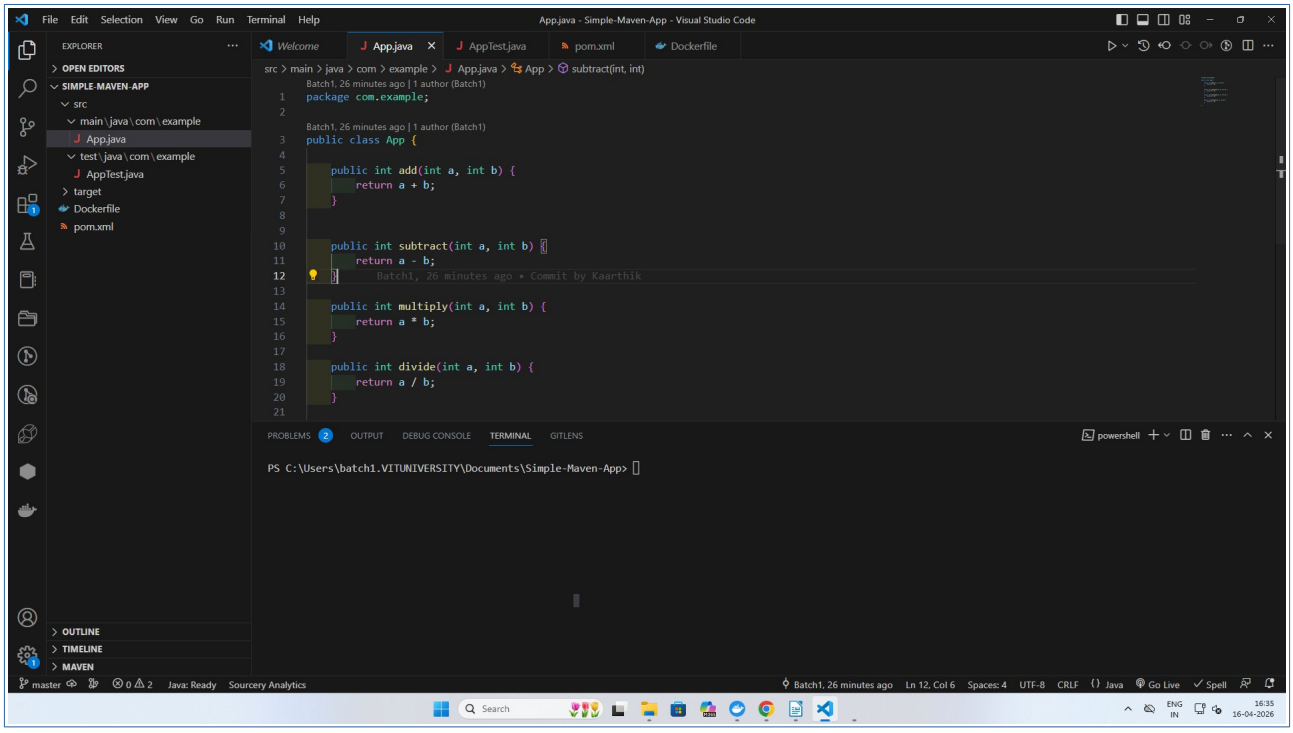
AGILE DEVELOPMENT PROCESS AND DEVOPS LAB
FINAL ASSESSMENT TEST
M KAARTHIK
23MIS0050

Q1. For the java application automate the continuous integration process using Jenkins. a. To implement the arithmetic operations.

Solutions:

1. Create a Maven-based application adhering to the standard Maven directory structure.



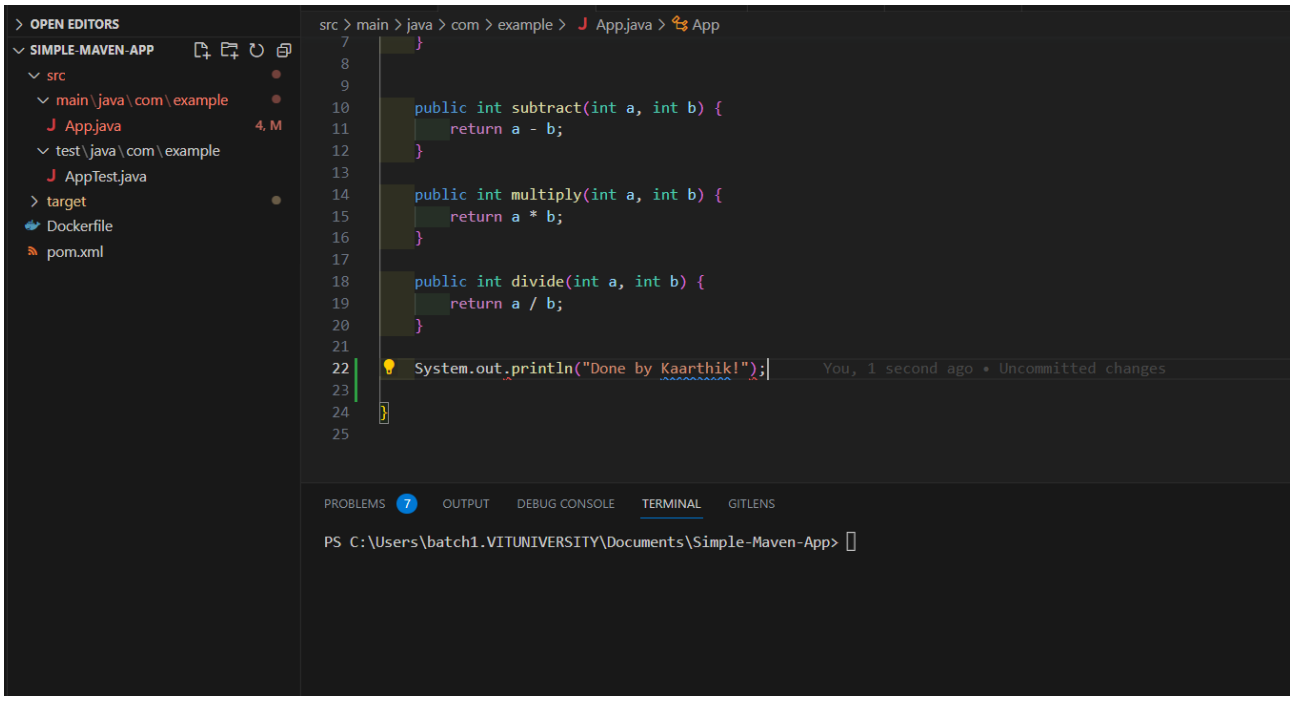
```
PROBLEMS 2   OUTPUT   DEBUG CONSOLE   TERMINAL   GITLENS                                   powershell

PS C:\Users\batch1.VITUNIVERSITY\Documents\Simple-Maven-App\src> cd ..
PS C:\Users\batch1.VITUNIVERSITY\Documents\Simple-Maven-App> mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] -------------------< com.example:simple-java-app >--------------------
[INFO] Building simple-java-app 1.0
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- clean:3.2.0:clean (default-clean) @ simple-java-app ---
[INFO] Deleting C:\Users\batch1.VITUNIVERSITY\Documents\Simple-Maven-App\target
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ simple-java-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\batch1.VITUNIVERSITY\Documents\Simple-Maven-App\src\main\resources
[INFO]
[INFO] --- compiler:3.10.1:compile (default-compile) @ simple-java-app ---
[INFO] Changes detected - recompiling the module!
[WARNING] File encoding has not been set, using platform encoding UTF-8, i.e. build is platform dependent!
[INFO] Compiling 1 source file to C:\Users\batch1.VITUNIVERSITY\Documents\Simple-Maven-App\target\classes
[INFO]
[INFO] --- resources:3.3.1:testResources (default-testResources) @ simple-java-app ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory C:\Users\batch1.VITUNIVERSITY\Documents\Simple-Maven-App\src\test\resources
[INFO]
[INFO] --- compiler:3.10.1:testCompile (default-testCompile) @ simple-java-app ---
[INFO] Running com.example.AppTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.077 s -- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  2.551 s
[INFO] Finished at: 2026-04-16T16:08:04+05:30
[INFO] ------------------------------------------------------------------------
```

2. Write the code in Visual Studio and push it to your GITHUB repository.

```
PROBLEMS 2   OUTPUT   DEBUG CONSOLE   TERMINAL   GITLENS                                   powershell

PS C:\Users\batch1.VITUNIVERSITY\Documents\Simple-Maven-App> git init
Initialized empty Git repository in C:/Users/batch1.VITUNIVERSITY/Documents/Simple-Maven-App/.git/
PS C:\Users\batch1.VITUNIVERSITY\Documents\Simple-Maven-App> git add .
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst', LF will be replaced by CRLF the next time Git tou
ches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst', LF will be replaced by CRLF the next time Git touch
es it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst', LF will be replaced by CRLF the next time
 Git touches it
warning: in the working copy of 'target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst', LF will be replaced by CRLF the next time G
it touches it
warning: in the working copy of 'target/surefire-reports/TEST-com.example.AppTest.xml', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\batch1.VITUNIVERSITY\Documents\Simple-Maven-App> git commit -m "Commit by Kaarthik
>> '^C
PS C:\Users\batch1.VITUNIVERSITY\Documents\Simple-Maven-App> git commit -m "Commit by Kaarthik"
[master (root-commit) 6055142] Commit by Kaarthik
 Committer: Batch1 <Batch1@VITUNIVERSITY.LOCAL>
After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 12 files changed, 164 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 pom.xml
 create mode 100644 src/main/java/com/example/App.java
 create mode 100644 src/test/java/com/example/AppTest.java
 create mode 100644 target/classes/com/example/App.class
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
 create mode 100644 target/surefire-reports/TEST-com.example.AppTest.xml
 create mode 100644 target/surefire-reports/com.example.AppTest.txt
 create mode 100644 target/test-classes/com/example/AppTest.class
PS C:\Users\batch1.VITUNIVERSITY\Documents\Simple-Maven-App> git remote add origin https://github.com/23MIS0050/Lab_FAT.git
PS C:\Users\batch1.VITUNIVERSITY\Documents\Simple-Maven-App> git push origin master
Enumerating objects: 37, done.
Counting objects: 100% (37/37), done.
Delta compression using up to 20 threads
```

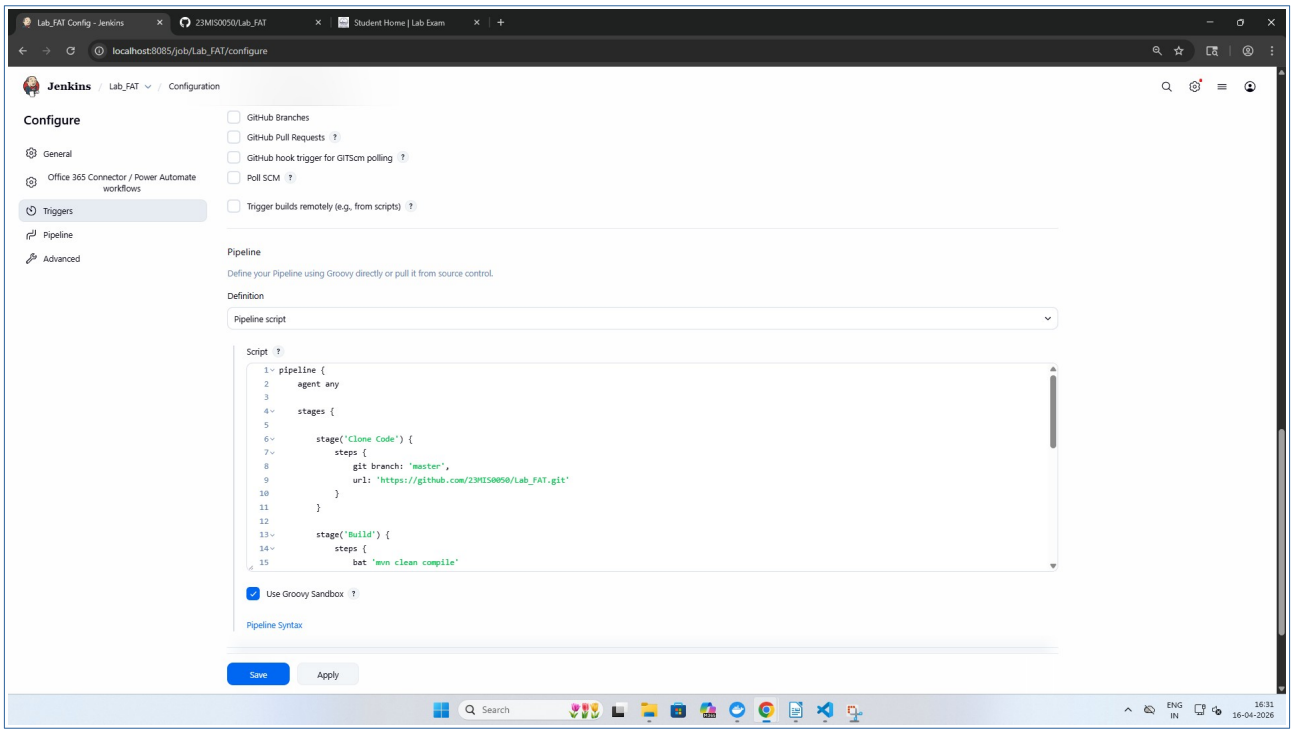
3 . Modify the code and commit the updates back in GITHUB.

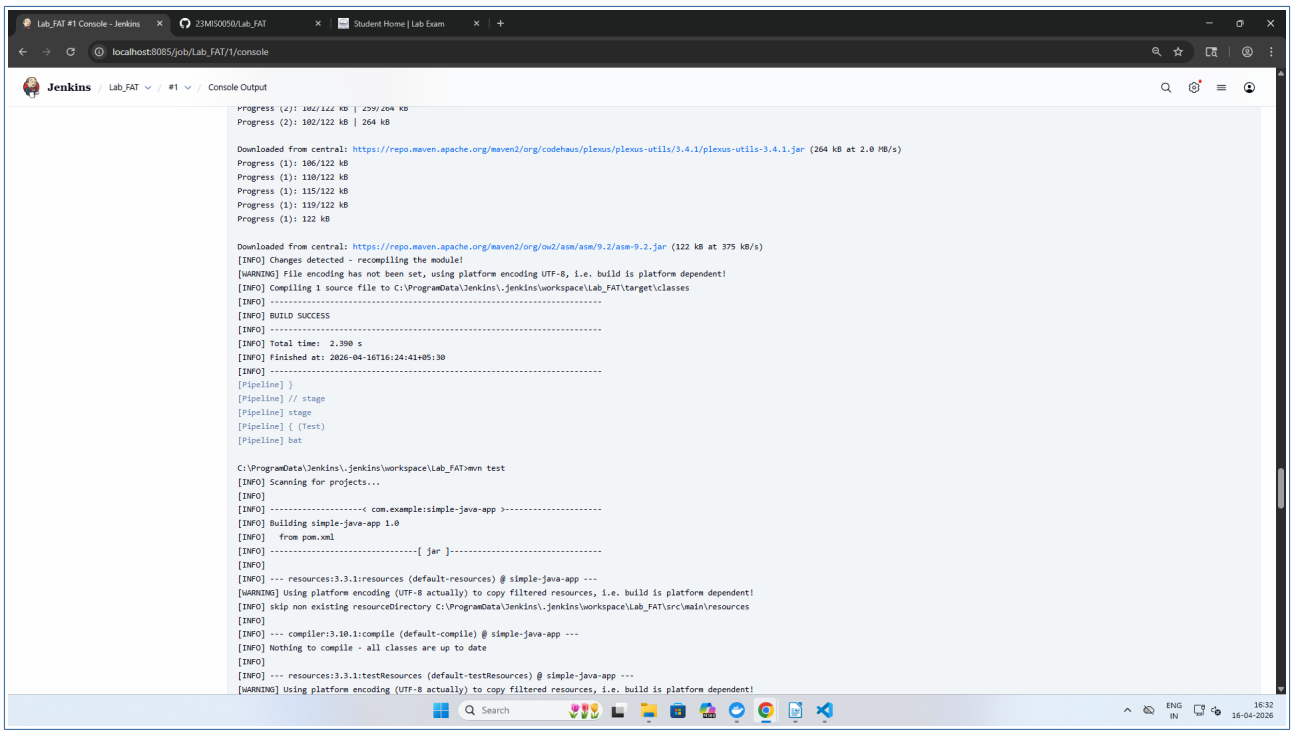
Modified the JavaFile



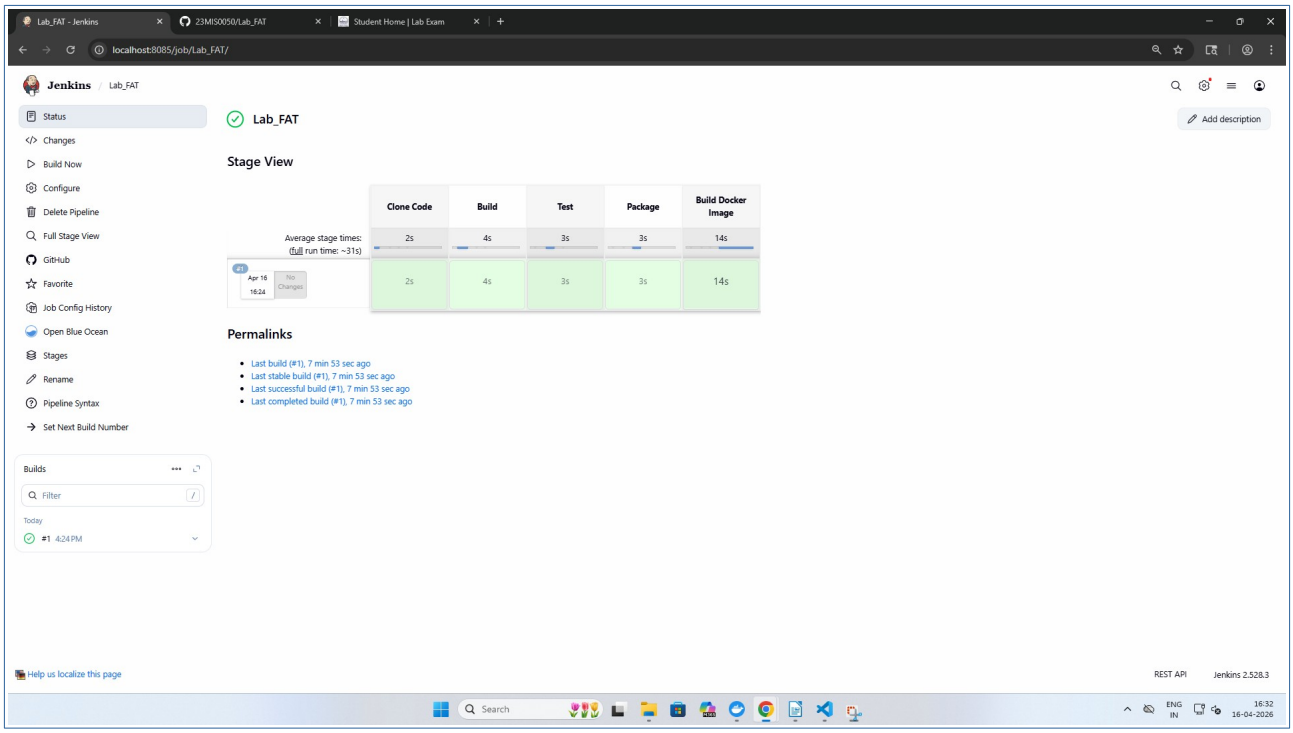
```
PROBLEMS 7   OUTPUT   DEBUG CONSOLE   TERMINAL   GITLENS                                   powershell

PS C:\Users\batch1.VITUNIVERSITY\Documents\Simple-Maven-App> git add .
PS C:\Users\batch1.VITUNIVERSITY\Documents\Simple-Maven-App> git commit -m "Java File Modified"
[master 21fcc4d] Java File Modified
 Committer: Batch1 <Batch1@VITUNIVERSITY.LOCAL>
Your name and email address were configured automatically based
on your username and hostname. Please check that they are accurate.
You can suppress this message by setting them explicitly. Run the
following command and follow the instructions in your editor to edit
your configuration file:

    git config --global --edit

After doing this, you may fix the identity used for this commit with:

    git commit --amend --reset-author

 1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\batch1.VITUNIVERSITY\Documents\Simple-Maven-App>


PS C:\Users\batch1.VITUNIVERSITY\Documents\Simple-Maven-App> git push origin master
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 20 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (8/8), 625 bytes | 125.00 KiB/s, done.
Total 8 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/23MIS0050/Lab_FAT.git
   68be28f..21fcc4d  master -> master
PS C:\Users\batch1.VITUNIVERSITY\Documents\Simple-Maven-App>
```

4. Using Jenkins configure the job as pipeline and create the stages for the Continuous Integration process.

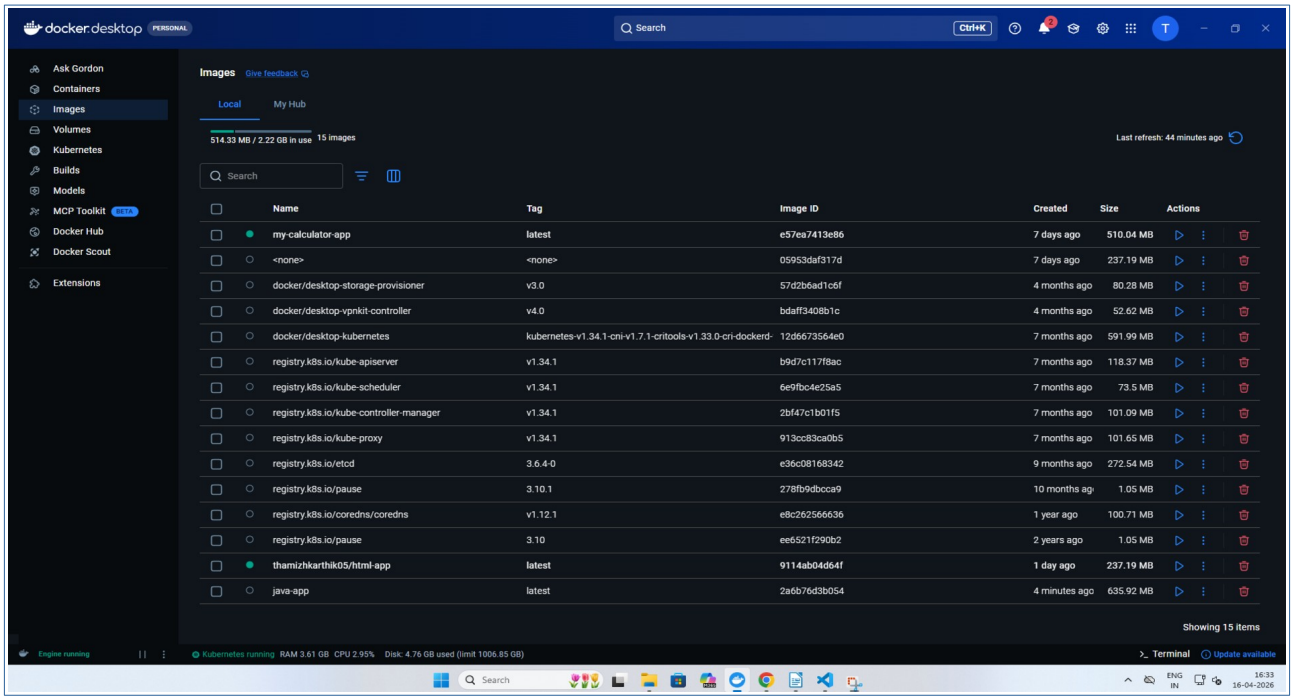




Test using JUNIT testing and building using Maven



Creation of Docker image



Pipeline Script:

```
pipeline {
    agent any

    stages {

        stage('Clone Code') {
            steps {
                git branch: 'master',
                url: 'https://github.com/23MIS0050/Lab_FAT.git'
            }
        }

        stage('Build') {
            steps {
                bat 'mvn clean compile'
            }
        }

        stage('Test') {
            steps {
                bat 'mvn test'
            }
        }
```

```
        stage('Package') {
            steps {
                bat 'mvn package'
            }
        }

        stage('Build Docker Image') {
            steps {
                bat 'docker build -t java-app .'
            }
        }
    }
}
```

GitHub Link : https://github.com/23MIS0050/Lab_FAT.git